GradeSense — Student Answer Script

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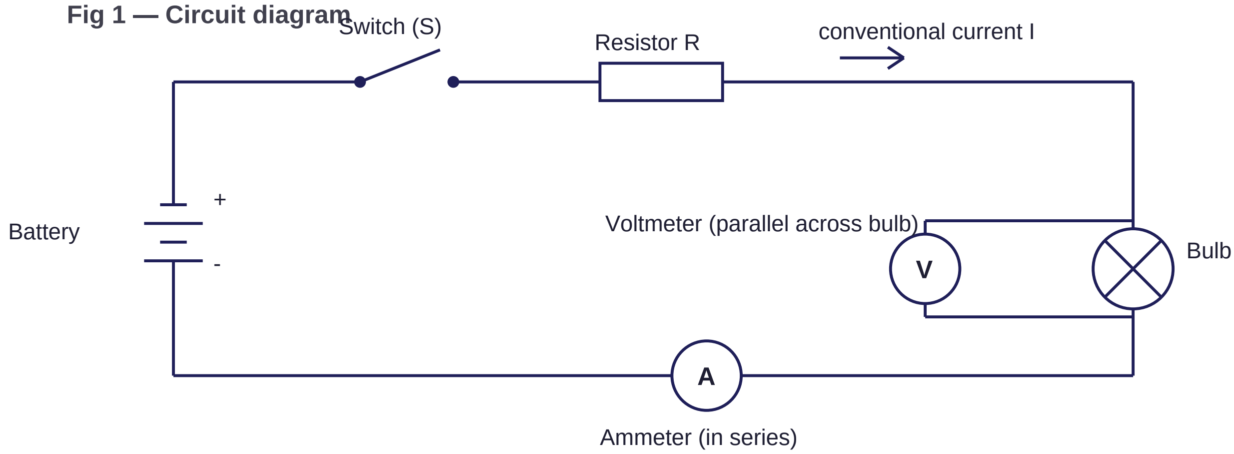
Question 1 — Science (5 marks)

A simple circuit is a closed loop through which current flows. The battery provides the potential difference and the switch completes or breaks the loop. The bulb, resistor, battery, switch and ammeter are connected in series, and the voltmeter is connected in parallel across the bulb because it measures the potential difference across it.

Current flows from the positive terminal of the battery around the external circuit to the negative terminal. When resistance increases the current decreases, since for a fixed voltage the current is inversely proportional to the resistance.

Fig 1 — Circuit diagram



Question 2 — English (5 marks)

Technology has made information much easier to reach, and that has clearly helped students who want to understand something outside class hours. A student can pause a video and replay it, which is not possible with a teacher in a full classroom.

But easy access also encourages copying. If the answer is available immediately, many students will take it and move on without learning anything from the question. Some people argue that this is no different from copying from a friend, which students always did, and that is a fair point, though the scale is much larger now.

Overall I think technology helps students who use it to understand and harms those who use it to finish.

Question 3 — Economics (5 marks)

I have plotted the demand and supply schedules with quantity on the horizontal axis and price on the vertical axis. Demand slopes downward and supply slopes upward, as expected.

Reading from the table, the equilibrium is at a price of Rs 40 and a quantity of 40 units, because that is where the quantity demanded settles. At the equilibrium price the market clears and there is no tendency for the price to move.

Below the equilibrium price the quantity demanded exceeds the quantity supplied, which creates a shortage and pushes the price up. Above it the quantity supplied exceeds the quantity demanded, which creates a surplus and pushes the price down.

If production costs rise, producers supply less at each price, so the supply curve shifts to the left and the new equilibrium has a higher price with a lower quantity.

Fig 2 — Demand and supply